

POC 03: Smart Building Management Platform

- Monitoring, Backup And Cost Optimization

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Role: **AWS DevOps Engineer**

Public-safe name: **Smart Building Management Platform**

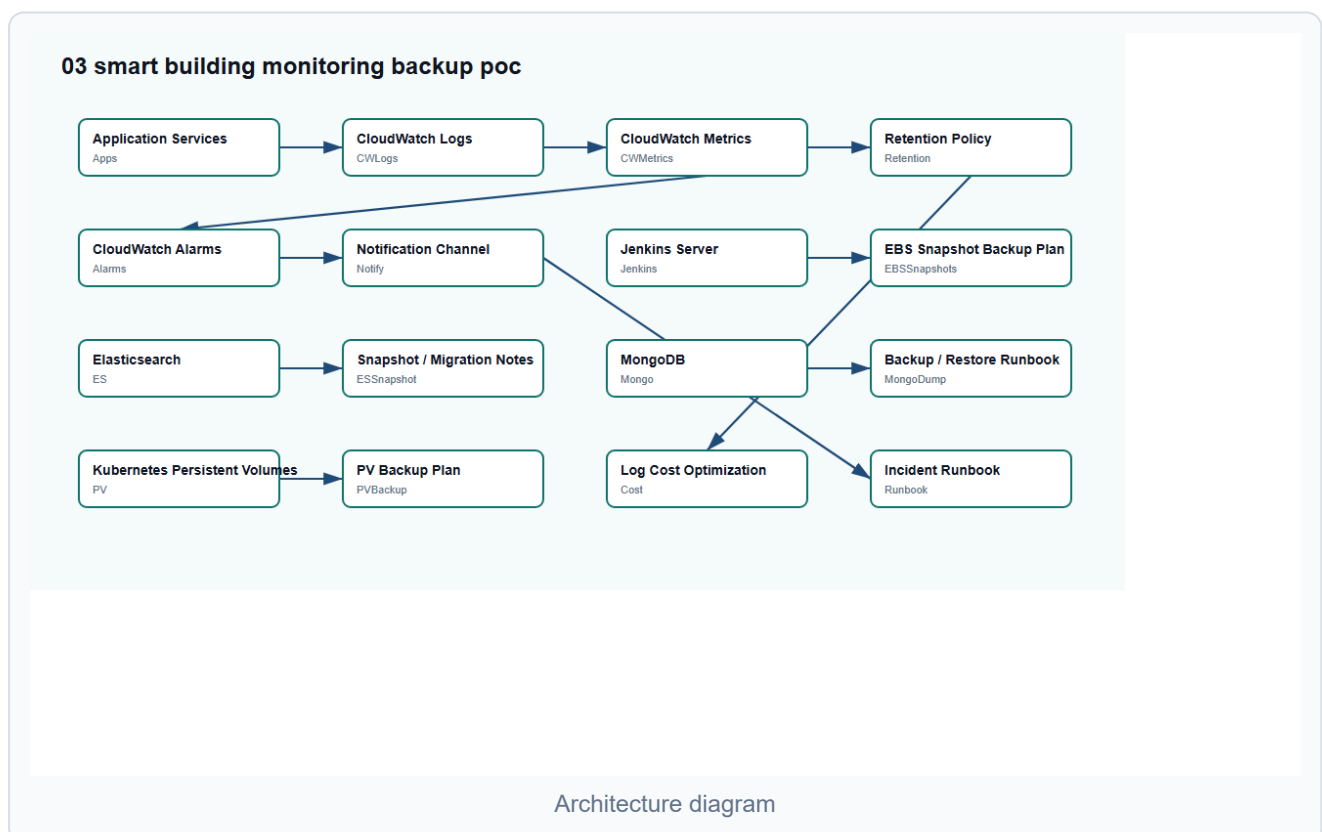
Confidentiality Note

This POC excludes private backup paths, repository names, environment values, credentials, internal dashboards, and production identifiers.

POC Objective

Build an operations blueprint for monitoring, alerting, backup planning, recovery documentation, Elasticsearch data handling, MongoDB backup/restore, Jenkins backup planning, and CloudWatch log cost control.

Operations Architecture Diagram



What I Worked On

- â€¢ Documented CloudWatch logging, monitoring, alerts, and notification practices.
- â€¢ Prepared Jenkins backup planning using EBS snapshot concepts.
- â€¢ Documented Elasticsearch backup/data migration and MongoDB backup/restore approach.
- â€¢ Added log retention and cost optimization guidance.

POC Implementation Scope

- â€¢ CloudWatch log retention example.
- â€¢ Alarm design for service errors, high CPU/memory, and failed jobs.
- â€¢ Backup runbooks for Jenkins, MongoDB, Elasticsearch, and persistent storage.
- â€¢ Recovery verification checklist.
- â€¢ Cost-control checklist for logs and backups.

Suggested Repository Structure

```
monitoring-backup-cost-poc/  
|-- README.md  
|-- cloudwatch/  
|   |-- alarms.md  
|   `-- log-retention.md  
|-- backup/  
|   |-- jenkins-backup.md  
|   |-- mongodb-backup-restore.md  
|   |-- elasticsearch-snapshot.md  
|   `-- pv-backup.md  
`-- runbooks/  
    |-- incident-response.md  
    `-- restore-verification.md
```

Validation Steps

- Review log retention and alarm thresholds.
- Run backup commands against dummy/local services where possible.
- Restore backup into a non-production target.
- Confirm row/document counts or service health after restore.
- Document restore outcome and improvement notes.

Expected Outcome

â€¢ Stronger operational readiness.

â€¢ Repeatable backup and restore documentation.

â€¢ Better CloudWatch log cost control.

â€¢ Clearer monitoring and incident response handover.

Technologies

AWS CloudWatch, EBS Snapshots, S3, EC2, Jenkins, Elasticsearch, MongoDB, Kubernetes Persistent Volumes, monitoring, alerting, backup automation, cost optimization.